APPLICATION OF COIR GEOTEXTILE FOR SLOPE STABILIZATION IN EMBANKMENT

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BACHELOR OF TECHNOLOGY

In

CIVIL ENGINEERING

of

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

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(ANISO 9001:2015 CERTIFIED INSTITUTION)

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December 2026

Department of Civil Engineering

VIDYA ACADEMY OF SCIENCE AND TECHNOLOGY

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(AN ISO 9001:2015 CERTIFED INSTITUTION)

certificate

This is to certify that the project report entitled **"APPLICATION OF COIR GEOTEXTILE FOR SLOPE STABILIZATION IN EMBANKMENT"** is a bona fide record of the work carried out by **Anitha N. K. (VAS23CE011), Anupa V. J. (LVAS23CE048),K Ashika Musthafa (VAS23CE023),**and **M Sidharth Anand (VAS23CE028)** at Vidya Academy of Science & Technology, Thalakkottukara, Thrissur-680501, in partial fulfilment of the requirementsfor the award of the degree of Bachelor of Technology in Civil Engineering of APJ Abdul Kalam Technological University during the academic year 2026-2027.

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I hereby undertake that the preliminary project work entitled **"APPLICATION OF COIR GEOTEXTILE FOR SLOPE STABILIZATION IN EMBANKMENT"** IS CARRIED OUT BY ME INDEPENDENTLY UNDER THE VALUABLE GUIDANCE OF **Dr Abhilasha P. S**., Head of Department and **Ms Nirmala Krishnan**, Asst Prof. of Dept. of Civil Engineering, Vidya Academy of Science and Technology, Thalakkottukara, Thrissur in partial fulfilment of the requirements for the award of degree of **Bachelor of Technology in Civil Engineering of APJ Abdul Kalam Technological University**, during the academic year 2026-2027.

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December 2026

Institution Vision

To be a premier institution in engineering education and research, nurturing innovative professionals for sustyainable societal development.

Institution Mission

M1: Provide quality engineering education that empowers students with strong technical competence and analytical problem-solving abilities.

M2: Promote research, innovation, entrepreneurship and industry collaboration to develop sustainable technological solutions.

M3: Develop ethical and socially responsible professionals with leadership qualities, lifelong learning abilities and a commitment to societal well-being.

Department vision

To excel in Civil Engineering education and research, moulding innovative and socially responsible professionals for sustainable development.

Department Mission

M1: Provide quality education in Civil Engineering that imparts technical knowledge and develops problem-solving skills.

M2: Promote research, innovation and entrepreneurship that contribute to sustainable infrastructure and societal development.

M3: Develop professionally competent Civil Engineers with ethical values, leadership qualities and commitment for societal well-being.

Department PEOs

PEO-1: Graduates will be able to apply knowledge of Civil Engineering principles, analytical skills, and modern engineering tools to provide innovative solutions for complex engineering problems.

PEO-2: Graduates will be able to excel in professional careers and higher studies by demonstrating technical competence, research aptitude, leadership qualities, and effective communication skills.

PEO-3: Graduates will be able to contribute to sustainable infrastructure development through innovation, entrepreneurship, and application of emerging technologies in Civil Engineering.

PEO-4: Graduates will be able to practice Civil Engineering with professional ethics, social responsibility, and a commitment to lifelong learning to address evolving societal and environmental challenges.

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PSO-2: Plan and execute construction and infrastructure projects using digital construction technologies, surveying and GIS tools, and project management practices for timely, cost-effective, and quality outcomes.

PSO-3: Innovate in emerging areas such as green building technologies, sustainable water resource management, and smart infrastructure systems to address societal needs.

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PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to the solution of complex engineering problems.

PO2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)

PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design systems/components/processes to be met identified needs with consideration for the public health and safety, whole-life cost net zero carbon, culture, society and environment as required. (WK5)

PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8)

PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognising their limitations to solve complex engineering problems. (WK2 and WK8)

PO6: The Engineer and The World: Analyse and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).

PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)

PO8: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

PO9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.

PO10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PO11: Life-Long Learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

(WK8)

Acknowledgement

During the course of our Project work, several persons collaborated directly and indirectly with us. Without their support it would be impossible for us to do our work. That is why we wish to dedicate this section to recognize their support.

First and foremost I thank God Almighty for His providence and for being the guiding light throughout the Project.

I would like to thank my guide **Dr Abhilasha P. S**., Head of CE Dept. and co-guide **Ms Nirmala Krishnan**, Asst. Prof., CE Dept. for providing critical inputs in the preparation of this report. I also thank all other faculty members in our department for their guidance.

I am thankful to **Dr Abhilasha P.S**., Head of Civil Department, and our Principal **Dr Sunitha C**., for their sole co-operation.

Finally, I would like to extend my sincere gratitude to friends who have always been helpful, in preparing and presenting the report and in the discussion following the presentation.

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Abstract

Embankment stability is an important consideration in road construction, particularly where weak soil, rainfall, waterlogging and inadequate drainage may contribute to deformation or slope failure. This project examines the application of coir geotextile as a natural reinforcement and erosion-control material for soil embankments. The project presentation identifies the NH 66 embankment failure at Kooriyad, Kerala, reported on 19 May 2025, as the event that motivated the study.

Coir geotextile is produced from coconut fibre and offers a locally available, biodegradable alternative to petroleum-based synthetic geotextiles. The proposed work combines site soil investigation, laboratory testing, comparison of reinforced and unreinforced soil, and slope-stability modelling using GeoStudio software. Soil tests listed in the presentation include sieve analysis, liquid-limit and plastic-limit tests, and direct shear testing. A Gradient Ratio Test (GRT) is also planned to examine the permeability-related behaviour of the soil–geotextile system.

The objectives are to investigate soil properties, evaluate embankment slope stability, apply coir geotextile, and develop a predictive model to simulate performance and service-life behaviour. The presentation does not provide completed laboratory values or final GeoStudio results; therefore, this report describes the proposed methodology and expected outcomes without presenting unverified numerical findings.

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CERTIFICATE

UNDERTAKING

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AKNOWLEDGEMENT

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Chapter 1

INTRODUCTION

Road embankments are built to elevate the road structure above the surrounding terrain. Their effectiveness is influenced by factors such as soil strength, design, drainage, compaction, and the relationship between the embankment and the underlying soil. When the soil becomes waterlogged, its effective strength can diminish, leading to increased pore-water pressures. Additionally, surface runoff and uncontrolled seepage may erode the slope and lead to instability.

The project presentation addresses the bund failure along the Kurumali River in Rappal, Thrissur, on June 9, 2026. It highlights the necessity of taking into account site conditions, drainage, and slope stability in the design of highway embankments. This project specifically aims to explore whether coir geotextile can enhance stability and control erosion in a soil embankment.

1.1 Geotextiles

Geotextiles are fabrics that allow water to pass through and are utilized alongside soil or various geotechnical substances. Based on their design and intended use, they can serve purposes such as separation, filtration, drainage, reinforcement, or erosion control. Typically, synthetic geotextiles are produced from polymers like polypropylene or polyester. In cases where sustainability and biodegradability are important, natural-fiber geotextiles made from materials like coir and jute are preferred.

1.2 Coir geotextile

Coir is derived from the husk of coconuts. Coir geotextiles can protect bare soil, reduce surface erosion, and offer reinforcement while maintaining beneficial tensile strength. Their porous design can also facilitate the growth of plants by safeguarding the soil surface and conserving moisture. The effectiveness of coir geotextiles is influenced by factors such as fiber quality, the method of weaving or construction, weight per unit area, installation practices, soil characteristics, and environmental conditions.



Fig 1 : Coir geotextile

1.3 Objectives

•To investigate the engineering properties of soil collected from the selected site.

•To evaluate the slope stability of the embankment.

•To study the application of coir geotextile in the soil embankment.

•To develop a predictive model in GeoStudio to simulate performance and service-life behavior.

1.4 Scope of the study

• Characterization of the site soil through suitable laboratory tests.

•Evaluation of soil strength, shear strength and settlement-related characteristics.

•Testing the permeability behavior of the soil–geotextile arrangement using a suitable method, including the proposed Gradient Ratio Test.

•Comparison of reinforced and unreinforced embankment conditions.

•Use of GeoStudio to assess slope stability under defined soil and embankment conditions.

•Consideration of environmental factors that may influence geotextile strength, durability and service life.

1.5 Sustainable development goals

The presentation links the project to four Sustainable Development Goals (SDGs):

SDG 9—Industry, Innovation and Infrastructure;

SDG 11—Sustainable Cities and Communities;

SDG 12—Responsible Consumption and Production

SDG 13—Climate Action. The connection arises from the investigation of a locally sourced natural material, the goal of improving infrastructure performance, and the consideration of sustainable material use and climate-related rainfall conditions.

1.6 Materials

•Soil samples collected from the selected site.

•Coir geotextile selected for the embankment application.

1.7 Equipments required

•Direct shear test apparatus.

•Sieve set for particle-size analysis.

•Cassagrande apparatus for liquid-limit testing.

•Gradient Ratio Test (GRT) apparatus.

1.8 Software required

GeoStudio software is identified for slope-stability analysis and development of the predictive model. The software analysis should use soil parameters obtained from the site investigation and clearly documented assumptions about the embankment geometry, groundwater or pore-water conditions, and geotextile representation.

1.9 Soil tests

1.9.1 Sieve analysis

Sieve analysis is proposed to determine particle-size distribution. A 1000 g soil sample taken and Cu, Cc are determined to find out the soil type from the size ranges below:

Gravel: 4.75–80 mm

sand: 0.075–4.75 mm

silt: 0.002–0.075 mm

clay: than 0.002 mm.

The mass retained on each sieve is used to calculate percentage retained, cumulative percentage retained and percentage finer. A particle-size distribution curve can then be plotted.

The effective particle size D10, coefficient of uniformity Cu = D60/D10, and coefficient of curvature Cc = $D30^2/(D60 \times D10)$ are to be obtained from the curve where the required values can be read.

1.9.2 Liquid limit

The liquid limit is determined using the Casagrande apparatus by recording container and soil masses, calculating water content, and relating water content to the number of blows. The liquid

limit is the water content corresponding to 25 blows. The flow index is calculated from the relationship between water-content change and the logarithm of the blow-count ratio.

1.9.3 Plastic limit

It is the water content at which the soil starts to crumble when rolled. The plastic limit is determined to find the consisrtancy index eventually.

1.9.4 Water content

This is determined just to find out the moisture content present in the sample

1.10 Future scope

•Conduct long-term studies of natural geotextile durability under different soil and moisture conditions.

•Extend slope-stability analyses to additional embankment geometries and groundwater conditions.

•Compare predicted results with field monitoring or physical model tests.

•Investigate how fibre quality, thickness, tensile properties and biodegradation affect service life.

•Study possible failure mechanisms and suitable remedial measures in future embankment investigations.

Chapter 2

LITERATURE REVIEW

Table 1:literature review

SL NO.	TITLE OF JOURNAL	TILTLE OF PAPER	NAME OF AUTHOR AND YEAR	METHOD/ TECHINIQUE	RESEARCH FINDINGS
1	scholarsmine.mst.edu	Case studies of application of coir and geotextiles for soil stabilization	K S Beena (2013)	Case Study	The coir is an effective, sustainable & economical geotextile
2	Textile Research Journal	Biodegradability study to develop longer life jute geotextiles for road applications	Mahuya Ghosh, Gauda Venkatappa Rao, Syamal kanti chakrabarthi, Supriya pal, uma shankar sharma(2019)	Laboratory Experime-ntal Study	Soil properties strongly influence durability, chemical treatments improves the service life of jute geotextiles
3	Inter national Journal of Engineering and Innovative Technology (IJEIT)	Application of coir geotextile in rural roads construction on BC soil sub grade	Kundan Meshram, S K Mital, P K Jain, P k Agarwal(2013)	Soil Reinforcement	the use of coir geotextiles significantly enhances the performance of Black Cotton soil
4	American Journal of EngineeringResearch(AJER)	Utilization of geotextile for soil stabilization	D A Ogundare, A O Familusi, A B Osunkunle (2018)	Laboratory testing	Improved subgrade strength can reduce pavement thickness, non

					woven geotextiles improve pavement performance
5	Inter national journal of Research inEngineering and Technology(IJERT)	Road embankment full scale investigation on soft soil with geotextile stabilization	Muhammad Arsyad, Indrasurya B Mochtar, Noor Endah Mochtar (2020)	Field trial and Numerical Analysis	Local soil stabilized can replace ordinary backfill material in piled embankment
6	Proceedings of Indian Geotechnical Conference(IGC-2009)	Coir geotextile as separation and filtration layer for low intensity road bases	K Rajagopal(2009)	Seperation and Fitration	The coir geotextiles are suitable for cost-effective field applications.
7	Journal of Natural Fibers	Durability study of coir geotextile embedded in different types of subgrade soil	M K Sayida, Sheela Evangeline(2020)	Laboratory Experiments	The durability of coir geotextile is highly dependent on soil type, with tensile strength reducing by 30-95 % 0ver 135 days due to biodegradation
8	Journal of Natural Fibers(Tailor &Francis)	Field evaluation of coir geotextile reinforced subgrade for low volume pavements	Anusudha Vishwanathan(2020)	Field performance evaluation	The effectiveness of the geotextiles depends on their physical properties

9	Geotextiles and Geomembranes(Science Direct)	Construction of unpaved rural road using jute-synthetic blended woven geotextile- A case study	G Basu, A N Roy, S K Bhattacharya, S K Ghosh (2009)	Laboratory Testing	The blended geotextile exhibits higher modulus, greater tensile strength, and better CBR puncture resistance than a conventional 100 % HDPE geotextile
10	Hydrology and Earth system (HESS)	The protective and attractive covering of a vegetated embankment using coir geotextile	S Vishnu das, H H G.Savenji (2006)	Controlled field Experiment	Geotextiles, when combined with local grass species, provide an effective, eco friendly solution
11	International Journal of Pavement Research and Technology	Performance of coir geotextile reinforced subgrade for low volume roads	V Anushitha, V Suntha, Samson Mathew(2020)	Laboratory Experiment	The bearing capacity of pavement increased when coir is placed at interference b/w subgrade and sub base
12	IJES	Highway Surface Drainage System & Problems of Water Logging In Road Section	Mr.Dipanjan Mukherjee (2014)	Laboratory Testing	It outlines structural solutions—such as proper surface sloping, inlets, open channels, and culverts—to efficiently manage runoff and preserve the

					road's intended lifespan.
13	IJESRT	Design Consideration for Flexible Pavement in Waterlogged Area	Mr. Harshal B. Patil, Prof. Suresh B. Sharma (2017)	Pavement Design Analysis	It highlights that stabilizing soft subgrade soil with 30% quarry dust or a cement-sand mixture maximizes soil bearing strength (CBR). Additionally, integrating subsurface drainage with plastic-coated aggregates and geogrid reinforcement significantly improves long-term pavement durability and moisture resistance.
14	SANDRP Website	NH66 in Kerala: Built Against Water logic, Designed to Collapse?	Sridhar Radhakrishnan	Case study	The SANDRP article reveals that the NH66 expansion in Kerala bypassed mandatory Environmental Impact Assessments by slicing the 644-km project into smaller sub-100 km segments, ignoring local monsoon hydrology and

					severe tree felling. By using 4–6 meter earthen embankments and grossly inadequate drainage culverts over reclaimed wetlands, the design acts like a dam blocking east-to-west water flow. Consequently, the highway causes widespread flooding, soil saturation, and structural collapses—such as the May 2025 Kooriyad failure

Chapter 3

METHODOLOGY

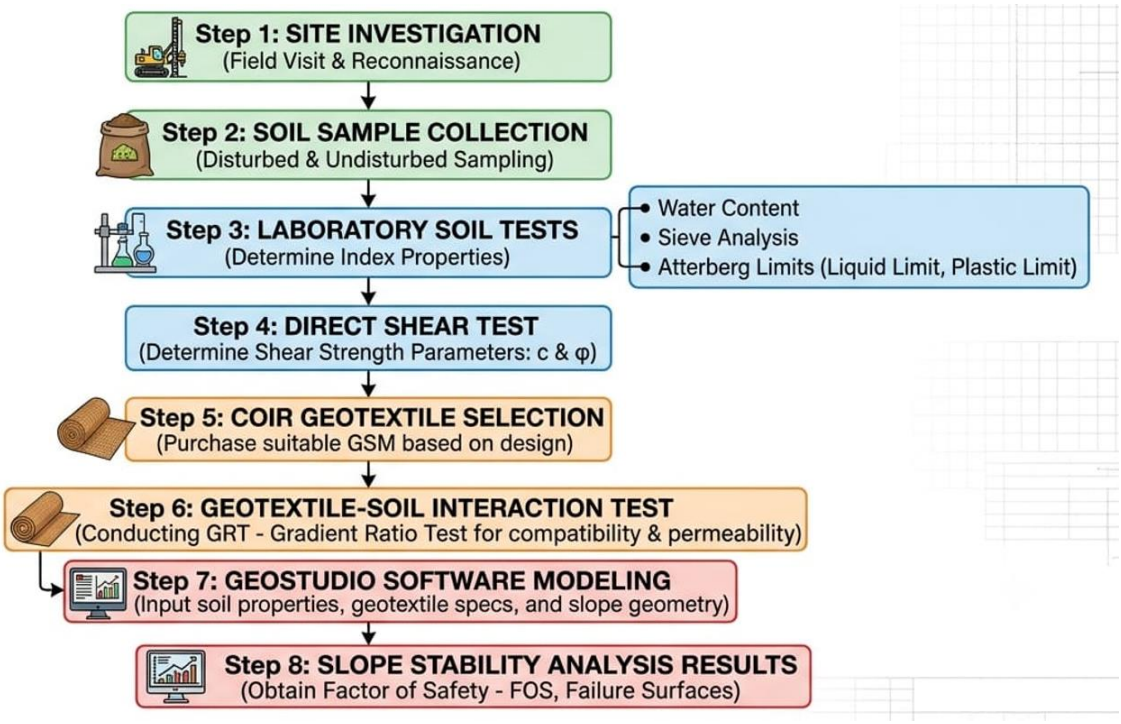


Fig 2: Flow chart of methodology

3.1 Site investigation

The site of bund failure in rappal is visited and discussed the event with the assistant engineer in which the location is under.

3.2 Soil sample collection

Three samples are collected from the failure slope. Three of the samples are collected in different heigths.

3.3 Laboratory soil tests

Water content, liquid limit, plastic limit, and sieve analysis tests are conducted on the collected soil samples

3.4 Direct shear test

The direct shear test is to determine soil shear-strength characteristics. Test results should be interpreted using the normal stress and measured shear response to estimate relevant strength parameters, such as cohesion and angle of internal friction, where these values have to be inputted in the GeoStudio software in future for slope analysis

3.5Coir geotextile selection

Coir geotextile of suitable GSM is used and the GSM is determined by using CBR test results.

3.6 Gradient Ratio Test

A Gradient Ratio Test is included in the planned work to evaluate the soil–geotextile interaction with respect to flow and potential clogging behavior. The test should be conducted with the selected soil and geotextile under a documented procedure. The resulting observations can be used to compare permeability-related behavior.

3.7 GeoStudio software modeling

A stimulative model is made in the GeoStudio software to analyze the slope stability by inputting the values obtained from the tests done on the collected soil.

3.8 Slope stability analysis results

Slope stability analysis result is obtained as an output from the software after analysing by inputting the different parameters which are obtained from the soil tests which are GRT, Direct shear, etc.

Chapter 4

CONCLUSION

This project aims to utilize coir geotextile for stabilizing slopes in a soil embankment, driven by worries regarding embankment performance and the failure of Kurumali river bund. The literature reviewed in the slides indicates that natural geotextiles can enhance soil reinforcement, separation, filtration, and erosion control, while also stressing the significance of material properties, biodegradation, and drainage.

The proposed research will integrate soil characterization, geotextile testing, and GeoStudio modeling. A key focus will be on comparing reinforced embankments with those that are unreinforced to assess the potential advantages of coir geotextile. As the presentation lacks finalized test results or numerical model data, the efficacy of the suggested solution will need to be confirmed through experimental and analytical methods.

Chapter 5

Project status completed in S7

5.1Site investigation

The site of bund failure in rappal is visited and discussed the event with the assistant engineer in which the location is under.



Fig 3: site investigation

5.2 Soil sample collection



Fig4: Collecting soil sample

Three samples are collected from the failure slope. Three of the samples are collected in different heigths.

5.3 Laboratory soil tests

5.3.1Water content

Table 2: Water content

SL NO	SAMPLE NO	1	2	3
1	Weight of container , w1	20	20	20
2	weight of container+wet soil,w2	50	40	45
3	weight of container + dry soil,w3	45	35	40
4	water content=(w2-w3)/(w3-w1) X100	40	33.33	25

5.3.2 Liquid limit

Determination no	1	2	3
Weigth of container,w0	20	20	20
Weigth of container+wet soil,w1	49	50	51
Weigth of container+dry soil,w2	40	40	40
Weigth of water (w1-w2)	9	10	11
Weigth of dry soil (w2-w0)	20	20	20
No of blows	30	28	27
Water content= (w1-w2)x100/(w2-w0)	45	50	55

Table 3:liquid limit

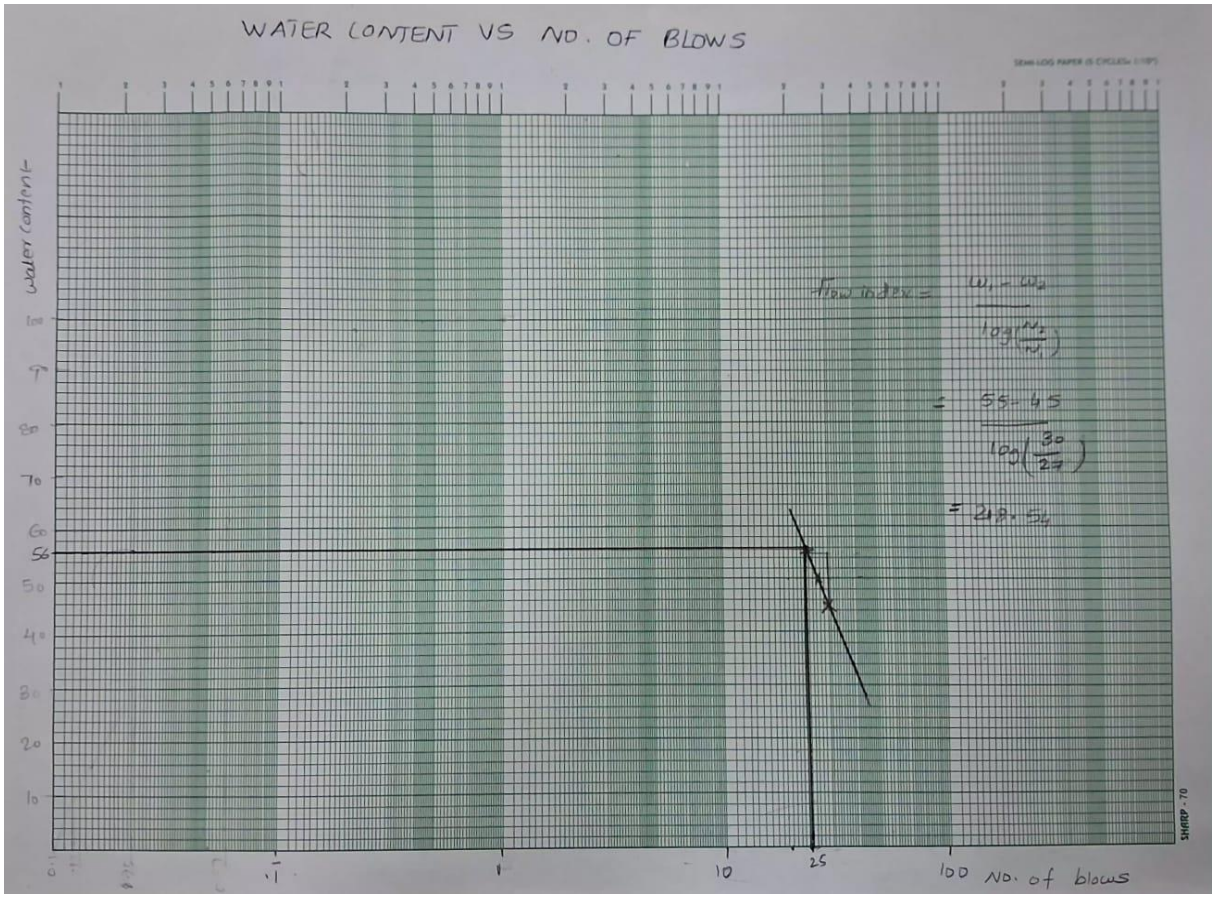


Fig 5: liquid limit graph

Liquid limit(water corresponding to 25 blows)= 56

Flow index= (w1-w2)/log(N2/N1)= (55-45)/log(30/27)= 218.54

5.3.3 Sieve analysis

Table 4: Sieve analysis

Sieve No	Sieve opening	Weigth of soil retained	Percent retained	Cumulative percent retained	Percent finer

1	1.18	700	70	70	30
2	0.600	240	24	94	6
3	0.425	40	4	98	2
4	0.300	10	1	99	1
5	0.075	10	1	100	0

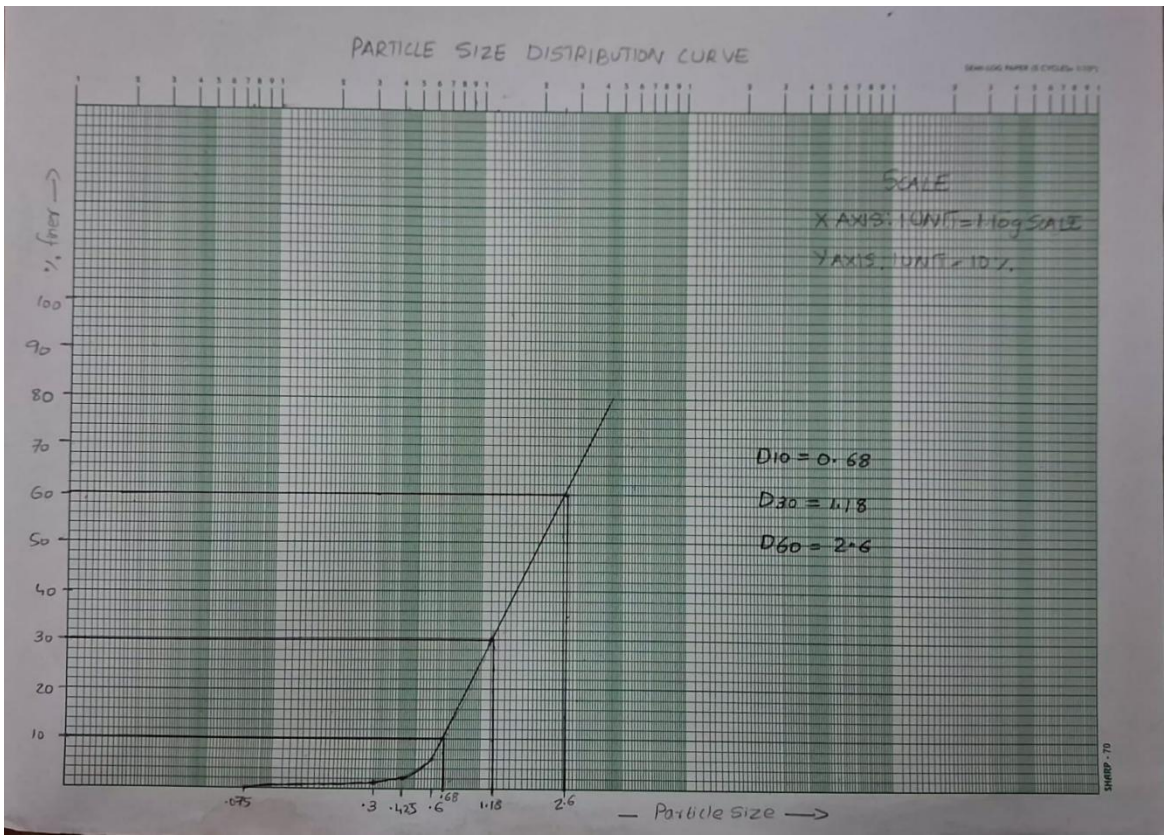


Fig6: Particle size curve

Effective particle size, D10 = 0.68
Coefficient of uniformity, Cu = D60/D10 = 3.82

Coefficient of curvature, Cc = D30^2/(D60xD10) = 0.787
The obtained value of Cu is 3.82, thus the soil is sandy soil.

Chapter 6

Project status to be completed in S8

Direct shear test

CBR test

Selecting suitable GSM Geotextile

GRT test

Slope stability analysis in GeoStudio Software

Study and evaluate the results

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